Sec 5.5 Nonhomogeneous eqs: Variation of parameters

Solving nonhomogeneous eq: $y'' + ay' + by = f(x)$

- if $f(x)$ is a polynomial, exponential, cos, sin, sinh, cosh, we can use Undetermined coefficients
- if $f(x)$ does not fit into these "known cases", or $f(x)$ "changes form*" with derivatives, then try variation of parameters (* ex: $f(x) = \ln x$, $f' = \frac{1}{x}$)

Setup: $y'' + ay' + by = f(x)$ $(f \neq 0)$

Ex: $y'' - y' - 2y = 3e^{2x}$

$$y_c = c_1 \underbrace{e^{-x}}_{y_1} + c_2 \underbrace{e^{2x}}_{y_2}$$

with undetermined coefficients, our trial $y_p = Ae^{2x}$, but we need to change this to avoid duplication with y_2, so we used

$$y_p = Axe^{2x}$$

Variation of params: "bootstrap" y_c into general solution $y = y_c + y_p$

try "solution" • $y = u_1(x)y_1 + u_2(x)y_2$, u_1, u_2 unknown functions

- $y' = u_1'y_1 + u_2'y_2 + u_1y_1' + u_2y_2'$
- (ex: $y = u_1e^{-x} + u_2e^{2x}$

$$y' = \underbrace{u_1'e^{-x} + u_2'e^{2x}} - u_1e^{-x} + 2u_2e^{2x}$$

★ assume that $u_1'e^{-x} + u_2'e^{2x} = 0$, since we are free to say. (so $u_1'y_1 + u_2'y_2 = 0$)

$$y' = -u_1 e^{-x} + 2u_2 e^{2x} \qquad \left(y = u_1 e^{-x} + u_2 e^{2x}\right)$$

$$\Rightarrow \; y'' = -u_1' e^{-x} + u_1 e^{-x} + \underbrace{2u_2' e^{2x}}_{2e^{2x} = y_2'} + \underbrace{4u_2 e^{2x}}_{4e^{2x} = y_2''}$$

Now subbing y, y', y'' into our equation

$$y'' - y' - 2y = 3e^{2x},$$

we the fact that $y_1 = e^{-x}$ and $y_2 = e^{2x}$ are solutions

to $y'' - y' - 2y = 0$,

to simplify and find that ...

$$u_1'(-e^{-x}) + u_2'(2e^{2x}) = 3e^{2x}$$

$$u_1' y_1' \quad + \quad u_2' y_2' = f(x).$$

So, recapping, in looking for u_1, u_2 to make a valid solution $y = u_1 y_1 + u_2 y_2$ for ODE $y'' - y' - 2y = 3e^{2x}$, we have assumed/found that:

$$\begin{cases} u_1' y_1 + u_2' y_2 = 0 & \leftarrow \text{assumed to simplify.} \\ u_1' y_1' + u_2' y_2' = f(x) & \longleftarrow \text{derived from } y'' - y' - 2y = f \end{cases}$$

(Ex)(cont:)

$$\begin{cases} u_1' \cdot (e^{-x}) + u_2' \cdot (e^{2x}) = 0 \\ u_1' \cdot (-e^{-x}) + u_2' \cdot (2e^{2x}) = 3e^{2x} \end{cases}$$

Goal: solve this for u_1', u_2', then integrate to get u_1, u_2.

$$\underbrace{\begin{bmatrix} e^{-x} & e^{2x} \\ -e^{-x} & 2e^{2x} \end{bmatrix} \begin{bmatrix} u_1' \\ u_2' \end{bmatrix}}_{A\underset{\sim}{x}} = \begin{bmatrix} 0 \\ 3e^{2x} \end{bmatrix}$$

$$= \underline{b}$$

If $\underline{A}^{-1}$ is defined, then we can solve $\underline{Ax} = \underline{b}$, b/c,

then $x = \underline{A^{-1}b}$.

$$\underline{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \Rightarrow \underline{A}^{-1} = \frac{1}{\det \underline{A}} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

assuming this $\neq 0$.

So, trying to solve

$$\overbrace{\begin{bmatrix} e^{-x} & e^{2x} \\ -e^{-x} & 2e^{2x} \end{bmatrix} \begin{bmatrix} u_1' \\ u_2' \end{bmatrix}}^{\underline{Ax}} = \overbrace{\begin{bmatrix} 0 \\ 3e^{2x} \end{bmatrix}}^{\underline{b}},$$

we would like to have $\underline{A}^{-1}$ here... so is det $\underline{A} \neq 0$?

$$\underbrace{\begin{vmatrix} e^{-x} & e^{2x} \\ -e^{-x} & 2e^{2x} \end{vmatrix}} = 2e^{x} + e^{x} = 3e^{x} \neq 0.$$

this determinant is $W(e^{-x}, e^{2x}) = W(y_1, y_2)$

so solution $\overset{x}{\begin{bmatrix} u_1' \\ u_2' \end{bmatrix}} = \overbrace{\frac{1}{3e^{x}} \begin{bmatrix} 2e^{2x} & -e^{2x} \\ e^{-x} & e^{-x} \end{bmatrix} \begin{bmatrix} 0 \\ 3e^{2x} \end{bmatrix}}^{A^{-1}\ b}$

$$= \frac{1}{3e^{x}} \begin{bmatrix} -3e^{4x} \\ 3e^{x} \end{bmatrix}$$

$$\begin{bmatrix} u_1' \\ u_2' \end{bmatrix} = \begin{bmatrix} -e^{3x} \\ 1 \end{bmatrix}$$

$$\Rightarrow u_1 = \int -e^{3x}\,dx = -\frac{1}{3}e^{3x} + c_1$$

$$\Rightarrow u_2 = \int 1\,dx = x + c_2$$

~~So, g~~

So general solution $y = u_1 y_1 + u_2 y_2$ for nonhomog.
$y'' - y' - 2y = 3e^{2x}$ is (where $y_1 = e^{-x}$, $y_2 = e^{2x}$)

$$y = \left(-\tfrac{1}{3}e^{3x} + c_1\right)e^{-x} + (x + c_2)e^{2x}$$
$$= c_1 e^{-x} + c_2 e^{2x} - \tfrac{1}{3}e^{2x} + xe^{2x}$$
$$= \underbrace{c_1 e^{-x} + \underbrace{\left(c_2 - \tfrac{1}{3}\right)}_{\to c_2} e^{2x}}_{\text{this is } y_c = c_1 y_1 + c_2 y_2} + \underbrace{xe^{2x}}_{\text{so this is our } y_p}$$

<u>Recap</u>: General method Summary

$$y'' + a(x)y' + b(x)y = f(x)$$ $f \neq 0$ (a, b don't need to be constant)

- Starting with* $y_c = c_1 y_1 + c_2 y_2$ (the solution for the homog. eqn $y'' + ay' + by = 0$) (* may be given, or may be found using char. eq.)
  want $u_1(x), u_2(x)$ to make general solution (of the nonhomog. eq.) as
  $$y = u_1 y_1 + u_2 y_2 \qquad (1)$$

- Assuming (by choice) that $u_1' y_1 + u_2' y_2 = 0$, we sub (2) into $y'' + ay' + by = f(x)$ and simplify to ultimately derive that we must have
  $$\begin{cases} u_1' y_1 + u_2' y_2 = 0 \\ u_1' y_1' + u_2' y_2' = f \end{cases}$$

$$\Rightarrow \begin{bmatrix} y_1 & y_2 \\ y_1' & y_2' \end{bmatrix} \begin{bmatrix} u_1' \\ u_2' \end{bmatrix} = \begin{bmatrix} 0 \\ f \end{bmatrix}$$
"$\underline{A}\underline{x} = \underline{b}$"

• Since $\det A = \det \begin{bmatrix} y_1 & y_2 \\ y_1' & y_2' \end{bmatrix} = W(y_1, y_2) \neq 0$;*

(* why? because to make complementary function $y_c = c_1 y_1 + c_2 y_2$ / general solution for $y'' + ay' + by = 0$, y_1, y_2 must be linearly indep $\Rightarrow W(y_1, y_2) \neq 0$)

we can solve the system $\underline{A}\underline{x} = \underline{b}$ via $\underline{x} = \underline{A}^{-1}\underline{b}$; specifically

$$\underbrace{\begin{bmatrix} u_1' \\ u_2' \end{bmatrix}}_{x} = \underbrace{\frac{1}{W}\begin{bmatrix} y_2' & -y_2 \\ y_1' & y_1 \end{bmatrix}}_{A^{-1}} \underbrace{\begin{bmatrix} 0 \\ f \end{bmatrix}}_{b} = \frac{1}{W}\begin{bmatrix} -y_2 f \\ y_1 f \end{bmatrix} = \begin{bmatrix} -y_2 f / W \\ y_1 f / W \end{bmatrix}$$

Thus:

$$\begin{cases} u_1' = \dfrac{-y_2 f}{W} \\ u_2' = \dfrac{y_1 f}{W} \end{cases} \Longrightarrow \begin{cases} u_1 = \displaystyle\int \frac{-y_2 f}{W}\, dx \\ u_2 = \displaystyle\int \frac{y_1 f}{W}\, dx \end{cases}$$

Then finally, make general solution

$$y = u_1 y_1 + u_2 y_2 \quad (= y_c + y_p)$$

Ex: $y'' + y = \sec x$

$\sec x$ not one of our "known/simple cases", so we'll use variation of params to get general solution

1) Find y_c (and y_1, y_2): char eq for $y''+y=0$ is

$r^2+1=0 \implies r = \pm i$, so

$$y_c = c_1 \underbrace{\cos(x)}_{y_1} + c_2 \underbrace{\sin(x)}_{y_2}$$

2) Variation of params: $y = u_1 y_1 + u_2 y_2 = u_1 \cos x + u_2 \sin x$

... ultimately find $\begin{cases} u_1' \cos x + u_2' \sin x = 0 \\ u_1' \cdot (-\sin x) + u_2' \cdot (\cos x) = \sec x \end{cases}$

... from this, we found that $\begin{bmatrix} u_1' \\ u_2' \end{bmatrix} = \begin{bmatrix} -y_2 f / W(y_1, y_2) \\ y_1 f / W(y_1, y_2) \end{bmatrix}$

$$W(y_1, y_2) = W(\cos x, \sin x) = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix} = 1$$

so $\begin{cases} u_1' = -\sin x \cdot \dfrac{1}{\cos x} / 1 = -\dfrac{\sin x}{\cos x} \\ u_2' = \cos x \cdot \dfrac{1}{\cos x} / 1 = 1 \end{cases}$

$$u_1 = \int \frac{\overbrace{-\sin x \, dx}^{du}}{\underbrace{\cos x}_{u}} = \ln|\cos x| + c_1$$

$$u_2 = \int 1 \, dx = x + c_2 .$$

$$y = u_1 y_1 + u_2 y_2 = (\cos x) \ln|\cos x| + c_1 \cos x + x \sin x + c_2 \sin x$$

$$= \underbrace{c_1 \cos x + c_2 \sin x}_{y_c} + \underbrace{x \sin x + \cos(x) \ln|\cos x|}_{y_p}$$